

Tutorium – Business Process Technologies and Management

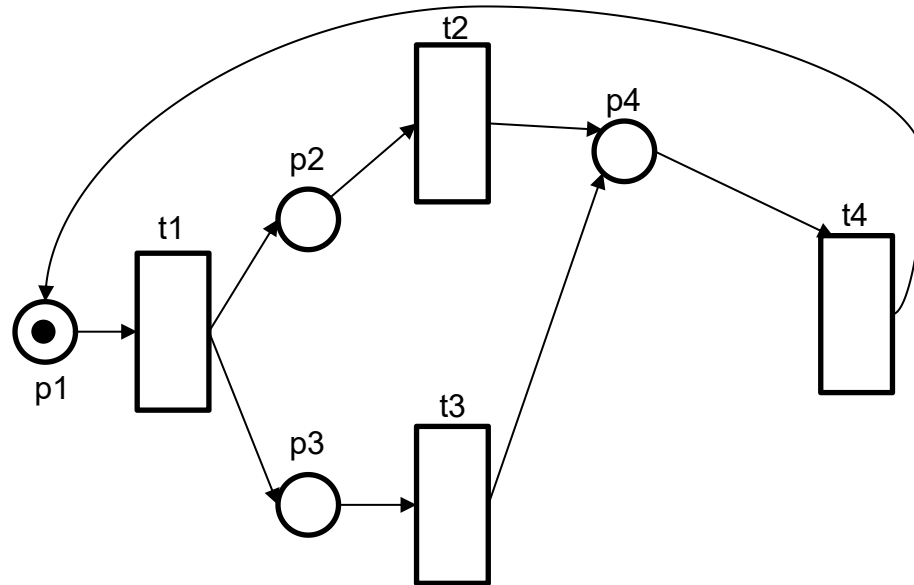
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Munich, 28. June 2022



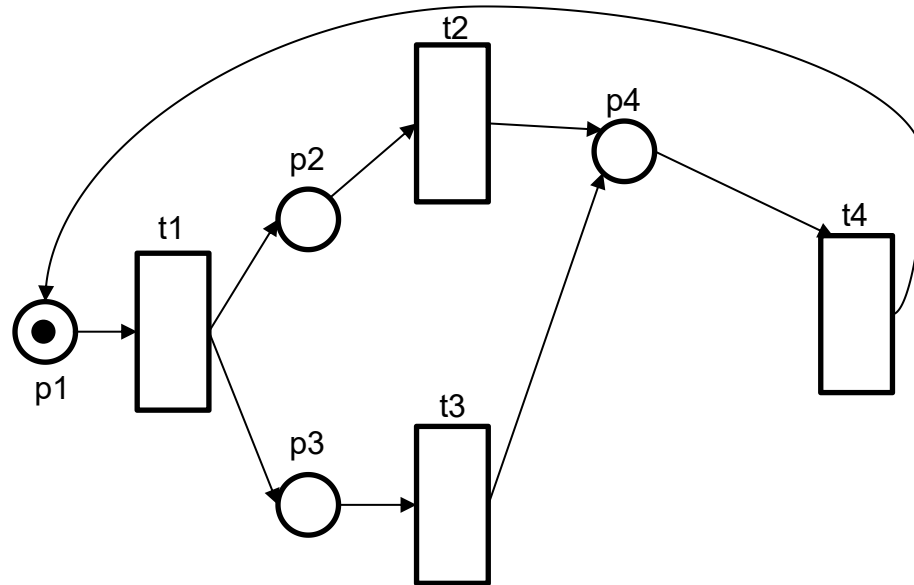
Petri Nets



Perform a reachability analysis for the following PT-NET

No.	P1	P2	P3	P4	Firing transition
0					
1					
2					
3					
4					
5					
6					
7					

Petri Nets



Perform a reachability analysis for the following PT-NET

No.	P1	P2	P3	P4	Firing transition
0	1	0	0	0	T1->m1
1	0	1	1	0	T2->m2 T3->m3
2	0	0	1	1	T3->M4 T4->M5
3	0	1	0	1	T2->M4
4	0	0	0	2	T4->M6
5					
6					
7					

Petri Nets

a) What type of Petri Net is this and why? Draw the Petri Net and Perform a Reachability analysis.

- $N = \{P, T, F, C, W, M_0\}$
- $P = \{p1, p2, p3, p4, p5, p6, p7\}$
- $T = \{t1, t2, t3, t4, t5\}$
- $F = \{\{p1, t1\}, \{p2, t2\}, \{t1, p2\}, \{p3, t3\}, \{t1, p3\}, \{t1, p4\}, \{p4, t4\}, \{p5, t4\}, \{t1, p5\}, \{t4, p7\}, \{p7, t5\}, \{t5, p1\}, \{t3, p6\}, \{t2, p6\}, \{p6, t5\}\}$
- $C = \{\}$
- $W = \{\{p1, t1, 2\}, \{t1, p2, 7\}, \{p2, t2, 7\}, \{t1, p5, 2\}, \{p5, t4, 2\}, \{p6, t5, 2\}\}$
- $M_0 = \{\{p1, 2\}\}$

b) Draw the Reachability graph.

c) What does C stand for and what is the value for p3?

d) What does W stand for and what is the value for $\{t1, p4\}$?

e) What is the minimal value for k-bounded and why?

f) Which Liveness does this net have and why?

Petri Nets

a) What type of Petri Net is this and why? Draw the Petri Net and Perform a Reachability analysis.

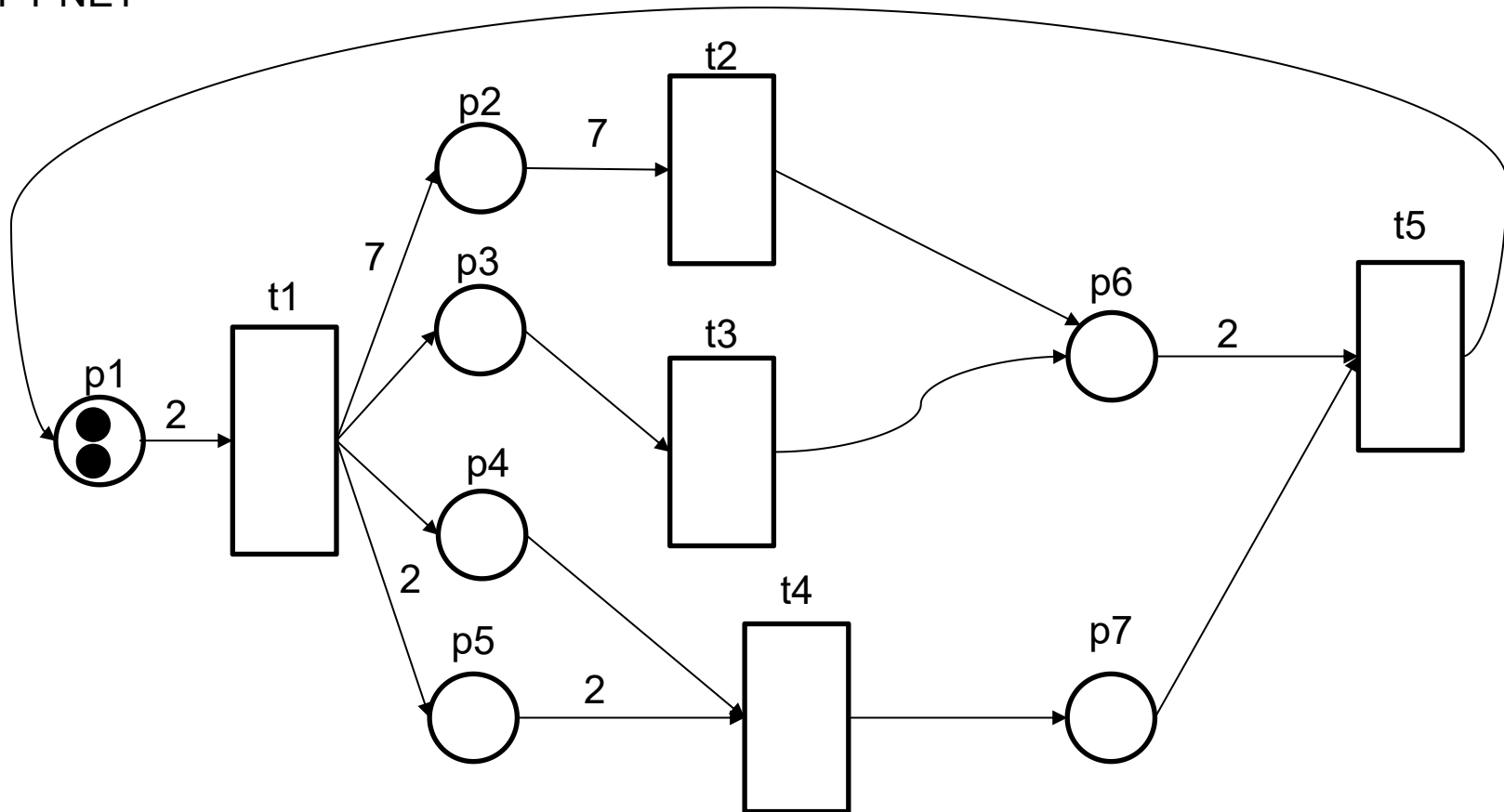
- $N = \{P, T, F, C, W, M_0\}$
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- $C = \{\}$
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Petri Nets

No.	P1	P2	P3	P4	P5	P6	P7	Firing transition

Petri Nets

PT-NET

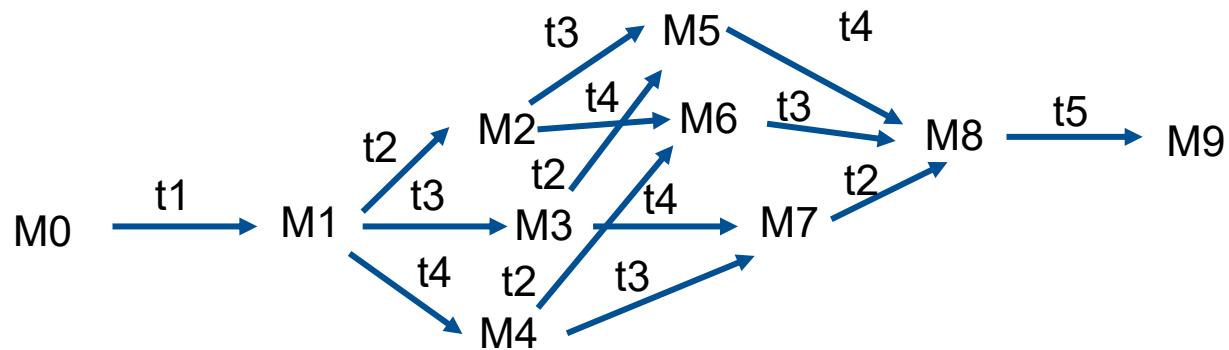


Petri Nets

No.	P1	P2	P3	P4	P5	P6	P7	Firing transition
0	2	0	0	0	0	0	0	T1->M1
1	0	7	1	1	2	0	0	T2->M2, T3-> M3, T4->M4
2	0	0	1	1	2	1	0	T3->M5, T4->M6
3	0	7	0	1	2	1	0	T2->M5 T4->M7
4	0	7	1	0	0	0	1	T2->M6 T3->M7
5	0	0	0	1	2	2	0	T4->M8
6	0	0	1	0	0	1	1	T3->M8
7	0	7	0	0	0	1	1	T2->M8
8	0	0	0	0	0	2	1	T5->M9
9	1	0	0	0	0	0	0	

b) Draw the Reachability graph

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c) What does C stand for and what is the value for p_3 ?

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Capacity value = Infinity

d) What does W stand for and what is the value for $\{t1, p4\}$?

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Weights , $\{t1, p4\}$ is 1

e) What is the minimal value for k -bounded and why?

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7 as this is the maximum value in the reachability analysis

f) Which Liveness does this net have and why?

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L1-Live as t_1 is L1-Live and all others are as well

- $L(M_0)$ all possible firing sequences from M_0

A petri net is L_k -Live if all transitions are L_k -Live with $k = 0,1,2,3,4$

A transition is

- L0-Live: the transition can never fire in any $L(M_0)$
- L1-Live: the transition can fire at least once for a sequence in $L(M_0)$
- L2-Live: the transition can fire a finite number of times for a sequence in $L(M_0)$
- L3-Live: the transition can fire an infinite number of times for a sequence in $L(M_0)$
- L4-Live: the transition can fire an infinite number of times for $[M_0 \rangle$

¹ Murata, Tadao. "Petri nets: Properties, analysis and applications." Proceedings of the IEEE 77.4 (1989): 541-580. <http://www2.ing.unipi.it/~a009435/issw/extra/murata.pdf>